

Sleep Disorder & Lifestyle Prediction Model

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Topic area: Health/Fitness

Dataset: Kaggle – Sleep and Lifestyle Study Dataset

Abstract:

Sleep disorders and poor sleep quality affect hundreds of millions of people worldwide, yet they frequently go undiagnosed until significant health consequences emerge. Conditions such as insomnia and sleep apnea are closely tied to lifestyle factors including physical activity, stress levels, occupation, and cardiovascular health, but traditional clinical screening relies on costly polysomnography tests and subjective self-reporting. Early, data-driven identification of at-risk individuals could substantially reduce this burden, making accurate predictive modeling in this space both timely and clinically valuable. Researchers have begun addressing this challenge through machine learning; one study published in the Journal of Medical Internet Research demonstrated that XGBoost combined with SHAP-based feature selection effectively predicted the risk of obstructive sleep apnea, comorbid insomnia, and insomnia without the need for polysomnography, while a separate team published in Diagnostics (2024) conducted a comprehensive comparison of regression and classification models on sleep disorder datasets and identified the best-performing learning strategies. Industry leaders are advancing this frontier further: Google has developed multimodal health models such as MedGemma and a Personal Health LLM built on Gemini, NVIDIA's Clara platform equips researchers with GPU-accelerated tools for medical imaging and health analytics, and Microsoft's InnerEye project applies deep learning to medical image analysis to improve clinical workflows. For this course project, I will build a sleep disorder classification model using the Sleep and Lifestyle Study Dataset from Kaggle, training and comparing a Random Forest Classifier and an XGBoost model across the dataset's patient features. These features include but are not limited to age, gender, occupation, sleep duration, sleep quality rating, physical activity level, stress level, BMI category, blood pressure, heart rate, and daily steps. The goal is to predict whether a patient has no sleep disorder, insomnia, or sleep apnea, evaluated using accuracy, F1-score, ROC-AUC, and per-class precision-recall metrics.

Potential Use Cases:

 = Interest/focus

- **Sleep Disorder Classification:** Predict whether a patient has no disorder, insomnia, or sleep apnea based on lifestyle and health metrics
- Early Detection Models: Identify individuals at risk for sleep disorders before symptoms become severe

- Feature Importance Analysis: Understand which lifestyle and physiological factors most strongly predict sleep disorder type
- **Stress-Sleep Correlation**: Map stress level and occupation patterns to specific sleep disorder categories
- Physical Activity Risk Assessment: Evaluate how daily step counts and activity levels correlate with sleep quality
- Public Health Modeling: Identify vulnerable populations (sedentary workers, high-stress occupations, individuals with elevated BMI)
- Clinical Decision Support: Assist healthcare providers in initial screening and differential diagnosis of sleep disorders

Classification Tasks:

- Multi-class classification (3 categories: No Disorder, Insomnia, Sleep Apnea)
- Binary classification (Disorder vs. No Disorder)
- Sleep quality score regression (predicting a continuous quality rating)
- Stress level prediction based on sleep and lifestyle inputs

What the Best Work is Doing:

- 1. Predicting the Risk of Sleep Disorders Using a Machine Learning–Based Simple Questionnaire (JMIR, 2023)**
 - a. What are they trying to solve?
 - i. The researchers aimed to develop an ML algorithm that could accurately predict the risk of obstructive sleep apnea (OSA), comorbid insomnia and sleep apnea (COMISA), and insomnia using a simple questionnaire, eliminating the need for expensive polysomnography testing.
 - b. What data did they use?
 - i. Data from two medical centers was used: 4,257 patients from Samsung Medical Center and 365 from Ewha Womans University Medical Center Seoul Hospital, with features selected via SHAP-based importance scoring.
 - c. What ML method did they use?
 - i. The team applied XGBoost (extreme gradient boosting) and used SHAP (Shapley Additive Explanations) values to identify and rank the most predictive features.
 - d. What's one important result?
 - i. The resulting tool, named SLEEPS, was deployed as a publicly accessible website, demonstrating that accurate sleep disorder risk prediction is achievable without costly clinical sleep tests.

2. Empirical Comparison of Sleep Disorder Prediction Using Machine Learning Techniques (ICRTC 2024, Springer)

- a. What are they trying to solve?
 - i. The authors investigated which machine learning algorithms are most effective at classifying sleep disorders, given the variability and complexity of lifestyle and health data.
- b. What data did they use?
 - i. The openly accessible Sleep Health and Lifestyle dataset from Kaggle was used, comprising 400 individual records categorized by sleep disorder status and featuring demographic, sleep, lifestyle, and physiological metrics.
- c. What ML method did they use?
 - i. Multiple algorithms were studied and compared: K-Nearest Neighbors, Naive Bayes, Support Vector Machine, Gradient Boosting Classifier, Decision Tree, and Random Forest, with outlier removal and categorical encoding during preprocessing.
- d. What's one important result?
 - i. The study concluded that ensemble methods such as Random Forest and Gradient Boosting consistently outperformed simpler models, underscoring the value of ensemble learning for structured health and lifestyle datasets.

A Dive Into 3 Industry Leaders:

Source: Top 10 AI Healthcare Companies – AI Magazine

Google

Sources: Google Health, Google PH-LLM

1. What product or technology are they building?

Google continues to develop AI-driven solutions to analyze patient data and health records for predictions and treatment recommendations. Relevant to sleep health, Google's models are increasingly being used to analyze wearable sensor data and longitudinal health trends.

2. How does ML help them?

Google has pioneered technologies such as MedGemma and TxGemma with the former offering multimodal medical text and image comprehension, and the latter being a collection of open models aimed at therapeutic development. These models can process diverse health inputs, including sleep-related data from devices like Fitbit.

3. What is one interesting detail you learned?

I was particularly intrigued by Google's Personal Health LLM (PH-LLM), a fine-tuned version of Gemini designed to provide personalized health insights and

recommendations, including sleep coaching. It can interpret longitudinal wearable data to offer individualized guidance.

NVIDIA

Source: NVIDIA Clara

1. What product or technology are they building?

NVIDIA's Clara platform is a suite of healthcare tools specifically designed for developers and researchers. Relevant tools include MONAI (Medical Open Network for AI), which supports medical imaging and physiological signal analysis that can be extended to sleep studies and EEG/ECG-based disorder detection.

2. How does ML help them?

Although NVIDIA does not directly provide patient care, their GPU-accelerated infrastructure and pre-trained models allow researchers to train complex classification models rapidly. This is directly applicable to training deep learning models on large-scale sleep datasets, such as polysomnography recordings.

3. What is one interesting detail you learned?

One interesting detail is that NVIDIA's DGX systems have been used to help train models for detecting anomalies in physiological signals, with potential applications in automated sleep stage classification and disorder screening — tasks directly relevant to this project.

Microsoft

Source: Project InnerEye – Democratizing Medical Imaging AI

1. What product or technology are they building?

Microsoft's Cloud for Healthcare is a collection of cloud services tailored for the health sector, incorporating AI tools that can analyze structured patient data, electronic health records, and sensor outputs — all relevant to sleep health monitoring.

2. How does ML help them?

Tools such as Azure Machine Learning and Microsoft Health Futures' InnerEye project employ CNNs and other deep learning approaches for medical data analysis. In the context of sleep, these tools could support the development of models that interpret physiological signals and automate disorder classification.

3. What is one interesting detail you learned?

An interesting detail was learning about the OSAIRIS project, which uses open-source software from InnerEye and Azure Machine Learning to help clinicians streamline workflows — reducing time from referral to treatment. A similar pipeline could be envisioned for sleep disorder screening, shortening the path from initial symptoms to diagnosis.

Questions to be Answered for this Project:

1. What dataset will you use (or what dataset type)?

I will be using the Sleep and Lifestyle Study Dataset from Kaggle (<https://www.kaggle.com/datasets/ayeshaimran1619/sleep-and-lifestyle-study>)

2. What is your input (features)?

The dataset contains features covering demographics, physiological metrics, and lifestyle habits. Key inputs include gender, age, occupation, sleep duration, sleep quality (rated 1–10), physical activity level (minutes/day), stress level (rated 1–10), BMI category, blood pressure, heart rate, and daily step count.

3. What is your output (target)?

Multi-class sleep disorder classification. The target is to predict which sleep disorder category a patient belongs to: No Disorder, Insomnia, or Sleep Apnea.

4. What is your algorithm (Classification, Regression, etc.)?

I will try both Classification using a Random Forest Classifier (RFC) and XGBoost.

5. Which models will you try?

I will try RFC as a strong ensemble baseline, and I am particularly interested in XGBoost for its accuracy and interpretability via SHAP values, as demonstrated in the literature.

Recommended Algorithms:

- Random Forest Classifier
- XGBoost / LightGBM
- Neural Networks
- Support Vector Machines
- Logistic Regression (baseline)

Evaluation Metrics:

- Accuracy
- F1-Score (macro/weighted)
- ROC-AUC
- Confusion Matrix Analysis
- Precision-Recall for each sleep disorder class